

Quasilocal Apparent-Horizon Spin-Function and Area Diagnostics

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Abstract

This whitepaper describes apparent-horizon diagnostics built from local two-surface geometry. The direct mass observable is the area, or equivalently the irreducible mass. Physical spin magnitude requires a separately normalized approximate-Killing or equivalent quasilocal rotation generator. The normal-bundle curvature Ω supplies a useful spin-function signal and coordinate-frame axis/proxy moments, but those moments are not by themselves a calibrated dimensionless spin vector and should not be used as input to the Christodoulou mass formula without an independent normalization.

1 Introduction and Literature Review

Black-hole diagnostics in numerical relativity are most useful when they are quasilocal: they are evaluated on a closed two-surface on a spatial slice rather than inferred from future null infinity or from the global event horizon. The surface considered here is an apparent horizon, more precisely an outer marginally trapped surface, on which the outgoing null expansion vanishes. Once this surface is known, its area gives an irreducible mass, its intrinsic and extrinsic geometry constrain possible angular momenta, and its proper circumferences provide stationary-Kerr consistency checks.

The diagnostic contract must keep four quantities separate. First, the horizon area gives the irreducible mass M_{irr} . Second, a physical spin magnitude requires a normalized quasilocal rotation generator, such as one obtained from an approximate Killing vector (AKV) construction. Third, moments of the normal-bundle curvature Ω provide spin-function and axis/proxy information in a chosen measurement frame, but are not calibrated dimensionless-spin components. Fourth, proper-circumference ratios are useful only as stationary-Kerr checks and are not general dynamical definitions of spin.

The theoretical foundation is the isolated- and dynamical-horizon program. Isolated horizons generalize black-hole mechanics to horizons that are locally in equilibrium even when the exterior is not stationary [1, 2]. Dreyer, Krishnan, Shoemaker, and Schnetter connected these ideas to numerical relativity by showing how mass and angular momentum can be extracted from apparent-horizon data when a rotational vector field is available [3]. Ashtekar and Krishnan extended the balance-law viewpoint to dynamical horizons [4, 5]. In standard 3+1 language, the associated angular momentum integral is closely related to Brown and York’s quasilocal surface integral [6]. The central difficulty is therefore not writing a surface integral, but choosing and normalizing the rotational vector field on a distorted, generally nonaxisymmetric surface.

The normal-bundle connection itself has a long history in horizon geometry. Hajicek’s null-horizon one-form and Damour’s membrane-paradigm horizon one-form give geometric and physical interpretations of the connection on the two-dimensional normal bundle [7, 8, 9, 10]. Its curvature is the imaginary part of the complex curvature of an oriented two-surface in the Penrose-Rindler

formalism [11, Sec. 4.14]. In vacuum and near-equilibrium settings, this curvature is closely tied to the horizon vorticity used in tendex-vortex descriptions of black-hole dynamics [12].

Approximate-symmetry constructions address the normalization problem. Cook and Whiting introduced a variational definition of approximate Killing vectors on topological two-spheres [13], and Beetle formulated a related eigenvalue problem suitable for numerical work [14]. Lovelace, Owen, Pfeiffer, and Chu used these ideas to control quasilocal spin in nearly extremal black-hole initial data [15]. Owen’s dynamical-horizon multipole analysis supplies the scalar-potential eigenproblem and current multipole framework used to extract normalized spin information from horizon data [16]. The SpECTRE code’s horizon diagnostics follow this lineage in their practical separation of spin function, AKV spin magnitude, Christodoulou mass, and normalized spin vector [17].

Spin direction is subtler than spin magnitude. Owen, Fox, Freiberg, and Jacques showed that common spin-axis definitions can show coordinate-dependent or gauge-sensitive nutation features, and they explicitly analyzed Ω -moment and centroid-corrected vorticity-dipole axes [18]. Thus an Ω -moment is valuable as a surface-field diagnostic and a frame-dependent tracking signal, but should not be promoted to a physical spin vector without independent normalization and validation.

The apparent-horizon setting imposes separate numerical constraints. Thornburg reviewed event and apparent horizon finding and developed fast direct methods for locating apparent horizons on numerical relativity grids [19, 20, 21]. Gundlach introduced pseudospectral horizon-finding ideas [22], and Lin and Novak developed a spectral apparent-horizon finder for three-dimensional simulations [23]. The Einstein Toolkit and its QuasiLocalMeasures module are useful points of comparison because they provide community infrastructure and surface-based calculations of quasilocal masses, spins, multipoles, and Killing-vector-based quantities on horizon surfaces [24, 25]. More recently, Etienne, Assumpcao, Werneck, and Tootle introduced BHaHAHA, a hyperbolic-relaxation apparent-horizon finder using multiresolution and over-relaxation acceleration [26]. The diagnostic described here attaches to such a horizon surface after convergence and evaluates area, expansion residuals, curvature moments, spin-function moments, and circumference-ratio checks.

2 Mathematical Formulation

All formulas use geometrized units $G = c = 1$. Let $(\Sigma, \gamma_{ij}, K_{ij})$ be a spatial slice with physical three-metric γ_{ij} and extrinsic curvature K_{ij} . Latin indices $i, j, \dots$ denote tensors on the spatial slice and uppercase indices $A, B, \dots$ denote tensors intrinsic to the horizon two-surface. The slice covariant derivative is D_i ; the surface covariant derivative compatible with the induced metric is D_A .

Let the apparent horizon $\mathcal{H}$ be a smooth topological two-sphere that is star-shaped with respect to the chosen angular coordinate center,

$$X^i(u^A) = (h(\theta, \phi), \theta, \phi), \quad F(r, \theta, \phi) = r - h(\theta, \phi) = 0, \quad (1)$$

where $u^A = (\theta, \phi)$ and h depends only on the angular coordinates. This representation assumes sufficient regularity at the coordinate poles and fails if the surface is not single-valued in the chosen radial coordinate. The tangent vectors are

$$e_A^i = \frac{\partial X^i}{\partial u^A}, \quad e_\theta^i = (h_\theta, 1, 0), \quad e_\phi^i = (h_\phi, 0, 1). \quad (2)$$

The outward unit normal in the slice is

$$s_i = \frac{\partial_i F}{\lambda}, \quad s^i = \gamma^{ij} s_j, \quad \lambda^2 = \gamma^{ij} \partial_i F \partial_j F. \quad (3)$$

The induced two-metric and area element are

$$q_{AB} = \gamma_{ij} e_A^i e_B^j, \quad q^{AC} q_{CB} = \delta_B^A, \quad dA = \sqrt{q} d\theta d\phi, \quad (4)$$

where $q = \det q_{AB}$. The orientation convention used below takes (θ, ϕ) to be a right-handed chart relative to the outward normal and sets

$$\varepsilon_{\theta\phi} = +\sqrt{q}, \quad \varepsilon^{\theta\phi} = +\frac{1}{\sqrt{q}}. \quad (5)$$

The extrinsic-curvature convention assumed in these formulas is the standard ADM/MTW/Wald numerical-relativity convention

$$K_{ij} = -\frac{1}{2} \mathcal{L}_n \gamma_{ij} = -\frac{1}{2\alpha} (\partial_t \gamma_{ij} - D_i \beta_j - D_j \beta_i), \quad (6)$$

where n^μ is the future-directed unit normal to the slice, α is the lapse, and β^i is the shift. Changing the orientation sign, changing s^i , or switching to the opposite convention for K_{ij} flips the reported signs of ω_A , Ω , and the spin moments. A positive-spin Kerr reference solution must therefore be used to lock the convention in any production workflow.

With this K_{ij} convention, the apparent-horizon condition is

$$\Theta = D_i s^i - K + K_{ij} s^i s^j = 0, \quad K = \gamma^{ij} K_{ij}. \quad (7)$$

The surface area and irreducible mass are

$$A = \oint_{\mathcal{H}} dA, \quad M_{\text{irr}} = \sqrt{\frac{A}{16\pi}}. \quad (8)$$

The displayed mass formula is the uncharged Christodoulou relation [27]; Christodoulou and Ruffini give the charged generalization [28]. It is a derived equilibrium relation and is meaningful here only after a dimensionful, physically normalized spin magnitude S has been supplied:

$$M_{\text{H}} = \sqrt{M_{\text{irr}}^2 + \frac{S^2}{4M_{\text{irr}}^2}}, \quad \chi = \frac{S}{M_{\text{H}}^2}. \quad (9)$$

A dimensionless spin vector is likewise $\chi^a = S^a/M_{\text{H}}^2$ only when S^a is a calibrated angular-momentum vector; its Euclidean component norm is a statement in the chosen measurement frame.

The spin-function construction begins with the tangential normal-bundle connection one-form

$$\omega_A = e_A^i K_{ij} s^j. \quad (10)$$

This expression already contains the tangential projection through e_A^i : if $q^i_j = \delta^i_j - s^i s_j$, then $\omega_A = e_A^p q_p^i K_{ij} s^j$. For tangential rotation fields φ^A , it is equivalent to the usual $(K_{ij} - K\gamma_{ij})$ quasilocal angular-momentum integrand because $\gamma_{ij} s^i \varphi^j = 0$ on $\mathcal{H}$. With the orientation above,

$$J[\varphi] = \frac{1}{8\pi} \oint_{\mathcal{H}} \omega_A \varphi^A dA. \quad (11)$$

Under a boost rescaling of the null normals, $\ell \mapsto e^\beta \ell$, $k \mapsto e^{-\beta} k$, the normal-bundle connection shifts by a gradient, $\omega_A \mapsto \omega_A + D_A \beta$, up to the chosen sign convention [5, 29]. The two-dimensional curl

$$\Omega = \varepsilon^{AB} D_A \omega_B \quad (12)$$

is therefore invariant under that boost-gauge shift. On a closed two-sphere, Hodge decomposition and $H^1(S^2) = 0$ imply that any smooth divergence-free tangential vector can be represented by a scalar potential ζ [30]:

$$\varphi^A = \varepsilon^{AB} D_B \zeta. \quad (13)$$

Because $\mathcal{H}$ is closed and $D_A \varepsilon^{AB} = 0$, integration by parts gives

$$J[\zeta] = \frac{1}{8\pi} \oint_{\mathcal{H}} \zeta \Omega \, dA. \quad (14)$$

This identity is the central reason that Ω is a natural spin-function integrand.

Physical AKV spin requires more structure. The scalar potentials ζ_α , $\alpha = 1, 2, 3$, are chosen so that $\varphi_\alpha^A = \varepsilon^{AB} D_B \zeta_\alpha$ are the best available rotational generators according to an approximate-Killing criterion. A common variational form minimizes

$$\mathcal{F}[\varphi] = \oint_{\mathcal{H}} \left(D_{(A} \varphi_{B)} - \frac{1}{2} q_{AB} D_C \varphi^C \right) \left(D^{(A} \varphi^{B)} - \frac{1}{2} q^{AB} D_C \varphi^C \right) dA \quad (15)$$

subject to a normalization condition [13, 14]. On the divergence-free scalar-potential subspace this variational problem becomes a fourth-order generalized eigenproblem for ζ , with equivalent forms differing by sign conventions for Δ , curvature notation, and normalization choices [14, 16]. Because the diagnostic described here does not solve that eigenproblem, the whitepaper does not promote any stand-alone partial differential equation as the implementation specification. A solver that supplies physical AKV modes must instead follow one of the cited exact formulations. Such solvers often introduce an auxiliary field $y \equiv \Delta \zeta$ to reduce the fourth-order problem to coupled second-order equations, but the operator and boundary/null-space treatment are part of the external AKV solve.

Mode selection, null-space handling, and normalization are part of the AKV spin calculation, not minor implementation details. Rescaling ζ_α rescales

$$S_\alpha = \frac{1}{8\pi} \oint_{\mathcal{H}} \zeta_\alpha \Omega \, dA, \quad S = \left(\sum_{\alpha=1}^3 S_\alpha^2 \right)^{1/2}. \quad (16)$$

Thus these equations define a physical spin magnitude only after the ζ_α have been externally normalized. This diagnostic adopts no AKV normalization internally. A downstream physical-spin workflow must state its normalization explicitly, for example a 2π -orbit normalization that reproduces the ordinary rotational generators in the axisymmetric Kerr or round-sphere limit [15, 16]. Without that external choice, $\int \zeta_\alpha \Omega \, dA$ is only a scale-dependent moment. The Ω -moment diagnostic described here does not by itself solve or normalize this AKV eigenproblem.

The Ω -moment axis is a separate, frame-dependent construction. Choose a measurement-frame embedding x^a , typically

$$x^a = (h \sin \theta \cos \phi, h \sin \theta \sin \phi, h \cos \theta). \quad (17)$$

This frame is not a diffeomorphism-invariant structure; it is the coordinate frame in which the axis or proxy is reported. Define the area centroid, the surface Ricci scalar $\mathcal{R}$, and the Ricci-weighted centroid correction

$$x_0^a = \frac{1}{A} \oint_{\mathcal{H}} x^a \, dA, \quad (18)$$

$$R_0 = \oint_{\mathcal{H}} \mathcal{R} \, dA, \quad (19)$$

$$x_R^a = \frac{1}{8\pi} \left(\oint_{\mathcal{H}} x^a \mathcal{R} \, dA - x_0^a R_0 \right). \quad (20)$$

The continuum value $R_0 = 8\pi$ follows from Gauss-Bonnet for a smooth topological two-sphere, but the integrated R_0 is the relevant discrete quantity for convergence monitoring. Owen et al.'s centroid-corrected vorticity-dipole direction is [18]

$$I^a = \oint_{\mathcal{H}} (x^a - x_0^a - x_R^a) \Omega \, dA, \quad n^a = \frac{I^a}{\sqrt{\delta_{bc} I^b I^c}}. \quad (21)$$

When a normalized spin magnitude S is independently known, one may report $S^a = S n^a$. The direction n^a inherits the coordinate-frame choice and is undefined when the dipole norm is numerically consistent with zero. A practical near-zero policy should compare $\sqrt{\delta_{ab} I^a I^b}$ with $\tau_\Omega L_{\mathcal{H}} \int |\Omega| \, dA$, where $L_{\mathcal{H}} = \sqrt{A/4\pi}$ and τ_Ω is a documented numerical tolerance. If this test fails, the axis should be reported as unavailable; output-only component fields may be set to zero or to a separately labeled fallback, but should not be interpreted as a physical spin direction.

The compact reported Ω -moment proxy is best written as a component triplet P_Ω^a . It is a diagnostic of record rather than a physical vector. For reproducibility, the proxy modes should be dimensionless measurement-frame $l = 1$ potentials $\hat{\zeta}_a$, reducing on a round sphere to

$$\hat{\zeta}_x = \sin \theta \cos \phi, \quad \hat{\zeta}_y = \sin \theta \sin \phi, \quad \hat{\zeta}_z = \cos \theta, \quad (22)$$

up to the same orientation sign fixed by the positive-Kerr convention. On a distorted surface, any supplied $\hat{\zeta}_a$ must document how it continues these round-sphere modes into the chosen measurement frame.

$$P_\Omega^a = \frac{1}{A} \left[\oint_{\mathcal{H}} x^a \Omega \, dA - \left(\frac{\oint_{\mathcal{H}} x^a \mathcal{R} \, dA}{\oint_{\mathcal{H}} \mathcal{R} \, dA} \right) \left(\oint_{\mathcal{H}} \hat{\zeta}_a \Omega \, dA \right) \right]. \quad (23)$$

Here $\hat{\zeta}_a$ is not the physical AKV potential ζ_a unless an external solver has explicitly made and normalized that identification. The proxy is useful for tracking a signed spin-function signal in the chosen frame, but it has no invariant magnitude without a calibration. Dimensionally, $\Omega \sim L^{-2}$, $dA \sim L^2$, and $x^a \sim L$, so $\oint x^a \Omega \, dA \sim L$. The second term in brackets has the same dimension only if $\hat{\zeta}_a$ is dimensionless: $(\oint x^a \mathcal{R} \, dA / \oint \mathcal{R} \, dA) \sim L$ and $\oint \hat{\zeta}_a \Omega \, dA \sim 1$. Dividing by $A \sim L^2$ then gives $P_\Omega^a \sim L^{-1}$. If the supplied proxy modes carry any other units, the formula is dimensionally inconsistent. Therefore P_Ω^a must not be identified with S , χ , χ^a , or inserted into M_H unless a separate Kerr or AKV calibration has established its units and interpretation.

The circumference-ratio diagnostic is a separate stationary-Kerr estimate. Let C_{eq} denote an equatorial proper circumference and C_p a polar proper circumference measured with respect to a known spin axis. For a stationary Kerr horizon with dimensionless spin $\chi = a/M$, Smarr's surface geometry gives the ratio used in numerical-relativity diagnostics [31, 32]:

$$\frac{C_p}{C_{\text{eq}}} = \frac{1 + \sqrt{1 - \chi^2}}{\pi} E \left(- \frac{\chi^2}{(1 + \sqrt{1 - \chi^2})^2} \right). \quad (24)$$

Here $E(m)$ is the complete elliptic integral of the second kind in Legendre parameter form, so m is the parameter k^2 , not the modulus k . The Kerr argument in this convention is negative. Inverting the relation gives a Kerr-like spin estimate only when the horizon is close to stationary and the chosen circumferences are aligned with the physical spin axis.

3 Numerical Methods

The surface is sampled on a cell-centered angular grid, $\theta_j = (j + \frac{1}{2})\Delta\theta$ and $\phi_k = (k + \frac{1}{2})\Delta\phi$, so no grid point lies exactly on a coordinate pole. The angular representation nevertheless assumes

a smooth spherical extension through the poles. Data supplied by the host evolution must be represented in a basis consistent with the (r, θ, ϕ) surface parameterization before the tangent vectors, normal, induced metric, and spin-function quantities are assembled. If the host data are naturally Cartesian, the metric, extrinsic curvature, and normal must be transformed before they are used in the surface tensor formulas.

The volume-to-surface data path is a two-stage interpolation problem. The fields needed by the diagnostic include the spatial metric, extrinsic curvature, conformal or ADM variables from which those tensors are built, and the surface derivatives required for Θ , $\mathcal{R}$, ω_A , and Ω . On a spherical source grid, angular data may first be interpolated with tensor-product Lagrange polynomials at fixed radius; values on the horizon are then obtained by Lagrange interpolation along radial spokes. If a stencil contains p_{int} points per direction, smooth data away from boundaries should show $O(\Delta^{p_{\text{int}}})$ interpolation error. The interpolation order, ghost-zone width, and basis of the interpolated tensor components must therefore be recorded with any diagnostic output.

Angular derivatives of h , surface fields, and supplied scalar potentials are computed with the configured even finite-difference order; fourth order is the default unless the driver chooses another even order. The pointwise geometric pipeline is to construct e_A^i , s^i , q_{AB} , $\sqrt{q}$, the surface connection, $\mathcal{R}$, ω_A , and $\Omega = \varepsilon^{AB} D_A \omega_B$. The same surface data give the expansion residual Θ , which should be monitored through surface-weighted L^2 and L^∞ norms.

All surface integrals are evaluated as weighted product quadratures:

$$\oint_{\mathcal{H}} f \, dA \approx \Delta\theta \, \Delta\phi \sum_{j,k} w_j w_k f_{jk} \sqrt{q_{jk}}. \quad (25)$$

The weights are midpoint-compatible. A single product-rule order is selected from the two angular grid dimensions: if both are divisible by eight, eighth-order weights are used; else if both are divisible by four, fourth-order weights are used; otherwise the uniform second-order midpoint rule is used. If one direction supports a higher-order stencil but the other does not, the lower common order is used for both. The azimuthal direction is periodic. In the polar direction, the use of the same stencil is justified only by regularity of the spherical representation and must be checked by convergence of $\int \Omega \, dA$, $\int \mathcal{R} \, dA$, and the corresponding moments.

For smooth star-shaped surfaces, the expected convergence rate of integrated diagnostics is controlled by the least accurate active ingredient:

$$p_{\text{eff}} = \min(p_{\text{fd}}, p_{\text{quad}}, p_{\text{int}}, p_{\text{AH}}),$$

where p_{fd} is the angular derivative order, p_{quad} the quadrature order, p_{int} the volume-to-surface interpolation order, and p_{AH} the observed convergence order of the horizon surface itself. With fourth-order angular derivatives, compatible fourth- or eighth-order quadrature, sufficiently high-order interpolation, and a converged smooth horizon, area-weighted residuals and Ω -moments should converge at approximately fourth order. Second-order midpoint fallback, pole nonregularity, horizon nonsmoothness, or unresolved interpolation lowers this target.

The principal reductions are

$$\begin{aligned} A, \quad \oint x^a \, dA, \quad \oint \mathcal{R} \, dA, \quad \oint x^a \mathcal{R} \, dA, \quad \oint \Omega \, dA, \\ \oint x^a \Omega \, dA, \quad \oint \zeta_\alpha \Omega \, dA, \quad \oint \hat{\zeta}_a \Omega \, dA, \quad \oint |\Omega| \, dA. \end{aligned} \quad (26)$$

These reductions directly give area, M_{irr} , centroids, curvature checks, Ω -moment axes, and proxy components. They do not by themselves produce calibrated AKV spin, dimensionless spin, or

Christodoulou mass; those require externally normalized spin potentials or an equivalent calibrated spin magnitude.

Coordinate-plane circumference ratios use the line elements

$$d\ell_\theta = \sqrt{q_{\theta\theta}} d\theta, \quad d\ell_\phi = \sqrt{q_{\phi\phi}} d\phi. \quad (27)$$

For a general-axis great circle with parameter α , both angular coordinates vary and the cross term is required:

$$d\ell = \left[q_{\theta\theta} \left(\frac{d\theta}{d\alpha} \right)^2 + 2q_{\theta\phi} \frac{d\theta}{d\alpha} \frac{d\phi}{d\alpha} + q_{\phi\phi} \left(\frac{d\phi}{d\alpha} \right)^2 \right]^{1/2} d\alpha. \quad (28)$$

Given a unit measurement-frame axis n^a , choose orthonormal vectors u^a and v^a with $u \cdot n = v \cdot n = u \cdot v = 0$. The equatorial coordinate great circle is represented on the unit sphere by

$$N_{\text{eq}}^a(\alpha) = u^a \cos \alpha + v^a \sin \alpha, \quad \alpha \in [0, 2\pi).$$

A polar great circle may be generated from

$$N_p^a(\alpha) = n^a \cos \alpha + u^a \sin \alpha.$$

The angles $\theta(\alpha) = \arccos N^z(\alpha)$ and $\phi(\alpha) = \text{atan2}(N^y(\alpha), N^x(\alpha))$ then determine the corresponding curve on the star-shaped surface $x^a = h(\theta, \phi) N^a$. Different choices of u^a rotate the polar meridian and should agree only when the surface is axisymmetric about n^a . The stationary-Kerr inversion should be reported as unavailable if the ratio is outside the Kerr-allowed range, if the spin axis is ambiguous, or if the horizon is too dynamical for a Kerr comparison to be meaningful.

Test	Expected continuum result	Interpretation if passed
Schwarzschild or time-symmetric data	$\Omega = 0$, $\int \Omega dA = 0$, and vanishing Ω -moments	Confirms nonspinning limit and near-zero handling
Kerr with known orientation	Correct area, M_{irr} , Ω sign, axis, and circumference ratio	Fixes sign conventions and stationary calibration checks
Gauss-Bonnet	$\int \mathcal{R} dA = 8\pi$ for a smooth topological sphere	Checks intrinsic curvature and quadrature consistency
Ω monopole	$\int \Omega dA = 0$ because Ω is a curl on a closed surface	Checks discrete curl, pole regularity, and quadrature
Mode rescaling	$\int \zeta_\alpha \Omega dA$ rescales with ζ_α	Demonstrates that AKV normalization is external
Frame translation/rotation	Directions transform covariantly where defined; proxy magnitudes remain frame-dependent	Clarifies coordinate-frame dependence
Near-zero spin	$\ I\ \leq \tau_\Omega L_{\mathcal{H}} \int \Omega dA$ gives unavailable axis	Protects nonspinning and weak-spin cases

The dominant numerical error sources are interpolation from volume data to the horizon, angular finite-difference error, polar regularity, quadrature error on distorted surfaces, normalization error in supplied spin potentials, and imperfect convergence of the apparent-horizon equation. The area and irreducible mass are the most direct outputs. All physical spin and horizon mass claims require a calibrated spin magnitude; Ω -moment quantities remain valuable tracking diagnostics only within the stated frame, convention, and validation limits. In addition to these numerical tests, the output schema should be audited so that proxy columns are labeled as Ω -moment proxies with their units, never as dimensionless χ^a .

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